28.0.1 UGC NET CSE | June 2009 | Part 2 | Question: 42

Which of the following is used for test data generation ?

A. White box
B. Black box
C. Boundary-value analysis
D. All of the above

ugcnetcse-june2009-paper2

Answer key

28.0.2 UGC NET CSE | June 2009 | Part 2 | Question: 39

Capability Maturity Model is meant for :

A. Product
B. Process
C. Product and Process
D. None of the above

ugcnetcse-june2009-paper2

Answer key

28.0.3 UGC NET CSE | June 2009 | Part 2 | Question: 40

In the light of software engineering software consists of :

A. Programs
B. Data
C. Documentation
D. All of the above

ugcnetcse-june2009-paper2

Answer key

28.0.4 UGC NET CSE | June 2009 | Part 2 | Question: 43

Reverse engineering is the process which deals with :

A. Size measurement
B. Cost measurement
C. Design recovery
D. All of the above

ugcnetcse-june2009-paper2

Answer key

28.0.5 UGC NET CSE | December 2018 | Part 2 | Question: 57

A legacy software system has 940 modules. The latest release required that 90 of these modules be changed. In addition, 40 new modules were added and 12 old modules were removed. Compute the software maturity index for the system.

A. 0.849
B. 0.524
C. 0.725
D. 0.923

ugcnetcse-dec2018-paper2

Answer key

28.0.6 UGC NET CSE | December 2004 | Part 2 | Question: 45

The Function Point (FP) metric is :

A. Calculated from user requirements
B. Calculated from Lines of code
C. Calculated from software's complexity assessment
D. None of the above

ugcnetcse-dec2004-paper2

Answer key ☟

28.0.7 UGC NET CSE | December 2004 | Part 2 | Question: 43

Reliability of software is dependent on :

A. Number of errors present in software
B. Documentation
C. Testing suites
D. Development Processes

ugcnetcse-dec2004-paper2

Answer key ☟

28.0.8 UGC NET CSE | December 2004 | Part 2 | Question: 42

Three essential components of a software project plan are :

A. Team structure, Quality assurance plans, Cost estimation
B. Cost estimation, Time estimation, Quality assurance plan
C. Cost estimation, Time estimation, Personnel estimation
D. Cost estimation, Personnel estimation, Team structure

ugcnetcse-dec2004-paper2

Answer key ☟

28.0.9 UGC NET CSE | December 2004 | Part 2 | Question: 41

The main objective of designing various modules of a software system is :

A. To decrease the cohesion and to increase the coupling
B. To increase the cohesion and to decrease the coupling
C. To increase the coupling only
D. To increase the cohesion only

ugcnetcse-dec2004-paper2

Answer key ☟

28.0.10 UGC NET CSE | December 2006 | Part 2 | Question: 45

'Abstraction' is ____________ step of Attribute in a software design.

A. First
B. Final
C. Last
D. Middle

ugcnetcse-dec2006-paper2

Answer key ☟

28.0.11 UGC NET CSE | December 2006 | Part 2 | Question: 44

Reliability of software is directly dependent on :

A. quality of the design.
B. number of errors present.

C. software engineer's experience.

D. user requirement.

ugcnetcse-dec2006-paper2

Answer key

28.0.12 UGC NET CSE | December 2006 | Part 2 | Question: 43

In a object oriented software design, 'Inheritance' is a kind of __________ .

A. relationship.
B. module.
C. testing.
D. optimization.

ugcnetcse-dec2006-paper2

Answer key

28.0.13 UGC NET CSE | December 2006 | Part 2 | Question: 42

Software Risk estimation involves following two tasks :

A. risk magnitude and risk impact.
B. risk probability and risk impact.
C. risk maintenance and risk impact.
D. risk development and risk impact.

ugcnetcse-dec2006-paper2

Answer key

28.0.14 UGC NET CSE | December 2006 | Part 2 | Question: 41

Software Cost Performance index (CPI) is given by :

A. $\frac{BCWP}{ACWP}$
B.
C. BCWP-ACWP
D. BCWP-BCWS

Where : BCWP stands for Budgeted Cost of Work Performed
BCWS stands for Budget Cost of Work Scheduled
ACWP stands for Actual Cost of Work Performed

ugcnetcse-dec2006-paper2

Answer key

28.0.15 UGC NET CSE | December 2006 | Part 2 | Question: 30

The LAPB frame structure and the frame structure of SDLC are :

A. Opposite
B. Identical
C. Reversed
D. Non-identical

ugcnetcse-dec2006-paper2

Answer key

28.0.16 UGC NET CSE | December 2006 | Part 2 | Question: 19

Which of the following tools is not required during system analysis phase of system development life cycle ?

A. Case tool
B. RAD tool
C. Reverse engineering
D. None of these

ugcnetcse-dec2006-paper2

Answer key

28.0.17 UGC NET CSE | December 2006 | Part 2 | Question: 16

Which possibility among the following is invalid in case of a Data Flow Diagram ?

A. A process having in-bound data flows more than out-bound data flows
B. A data flow between two processes
C. A data flow between two data stores
D. A data store having more than one in-bound data flows

ugcnetcse-dec2006-paper2

Answer key

28.0.18 UGC NET CSE | December 2007 | Part 2 | Question: 45

The approach to software testing is to design test cases to :

A. break the software
B. understand the software
C. analyse the design of sub processes in the software
D. analyze the output of the software

ugcnetcse-dec2007-paper2

Answer key

28.0.19 UGC NET CSE | December 2007 | Part 2 | Question: 44

System development cost estimation with use-cases is problematic because :

A. of paucity of examples
B. the data can be totally incorrect
C. the expertise and resource available are not used
D. the problem is being over simplified

ugcnetcse-dec2007-paper2

Answer key

28.0.20 UGC NET CSE | December 2007 | Part 2 | Question: 42

Function point metric of a software also depends on the :

A. number of function needed
B. number of final users of the software
C. number of external inputs and outputs
D. time required for one set of output from a set of input data

ugcnetcse-dec2007-paper2

Answer key

28.0.21 UGC NET CSE | December 2007 | Part 2 | Question: 41

A major defect in water fall model in software development is that :

A. the documentation is difficult
B. a blunder at any stage can be disastrous
C. a trial version is available only at the end of the project
D. the maintenance of the software is difficult

ugcnetcse-dec2007-paper2

Answer key

28.1 Adaptive Maintenance (1)

28.1.1 Adaptive Maintenance: UGC NET CSE | June 2019 | Part 2 | Question: 58

Software products need adaptive maintenance for which of the following reasons?

A. To rectify bugs observed while the system is in use
B. When the customers need the product to run on new platform
C. To support the new features that users want it to support
D. To overcome wear and tear caused by the repeated use of the software

ugcnetcse-june2019-paper2 adaptive-maintenance

Answer key

28.2 Artificial Intelligence (2)

28.2.1 Artificial Intelligence: UGC NET CSE | December 2012 | Part 3 | Question: 74

A* algorithm is guaranteed to find an optimal solution if

A. h' is always 0
B. g is always 1
C. h' never overestimates h
D. h' never underestimates h

ugcnetcse-dec2012-paper3 artificial-intelligence

Answer key

28.2.2 Artificial Intelligence: UGC NET CSE | June 2012 | Part 3 | Question: 23

Which of the following prolog programs correctly implement "if G succeeds then execute goal P else execute goal θ"?

A. if-else (G, P, θ):-!,call(G), call(P). if-else (G,P, θ) :- call(θ).
B. if-else (G, P, θ):-call(G), !, call(P). if-else (G,P, θ) :- call(θ).
C. if-else (G, P, θ):-call(G), call(P), !. if-else (G,P, θ) :- call(θ).
D. All of the above

ugcnetcse-june2012-paper3 artificial-intelligence

Answer key

28.3 Assignment Problem (1)

28.3.1 Assignment Problem: UGC NET CSE | June 2012 | Part 3 | Question: 62

The optimal solution of the following assignment problem using Hungarian method is

	I	II	III	IV
A	8	26	17	11
B	13	28	4	26
C	38	19	18	15
D	19	26	24	10

A. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
B. (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
C. (A)-(I), (B)-(III), (C)-(IV), (D)-(II)
D. (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

ugcnetcse-june2012-paper3 optimization assignment-problem

Answer key

28.4 CRC Polynomial (1)

28.4.1 CRC Polynomial: UGC NET CSE | December 2015 | Part 3 | Question: 31

In CRC based design, a CRC Team consists of

i. one or two users representatives
ii. several programmers
iii. project co-ordinators
iv. one or two system analysts

A. i and iii
B. i, ii, iii and iv
C. i, iii, and iv
D. i, ii, and iv

ugcnetcse-dec2015-paper3 is&software-engineering crc-polynomial

Answer key

28.5 Cmm Model (3)

28.5.1 Cmm Model: UGC NET CSE | December 2014 | Part 2 | Question: 42

Requirement Development, Organizational Process Focus, Organizational Training, Risk Management and Integrated Supplier Management are process areas required to achieve maturity level

A. Performed
B. Managed
C. Defined
D. Optimized

ugcnetcse-dec2014-paper2 is&software-engineering cmm-model

Answer key

28.5.2 Cmm Model: UGC NET CSE | July 2016 | Part 3 | Question: 47

Which of the following sets represent five stages defined by Capability Maturity Model (CMM) in increasing order of maturity?

A. Initial, Defined, Repeatable, Managed, Optimized
B. Initial, Repeatable, Defined, Managed, Optimized
C. Initial, Defined, Managed, Repeatable, Optimized
D. Initial, Repeatable, Managed, Defined, Optimized

ugcnetcse-july2016-paper3 is&software-engineering cmm-model

Answer key

28.5.3 Cmm Model: UGC NET CSE | June 2014 | Part 3 | Question: 21

Which one of the following is not a key process area in CMM level 5?

A. Defect prevention
B. Process change management
C. Software product engineering
D. Technology change management

ugcnetjune2014iii is&software-engineering cmm-model

Answer key

28.6 Cmmi (1)

28.6.1 Cmmi: UGC NET CSE | July 2018 | Part 2 | Question: 15

Match the 5 CMM Maturity levels/CMMI staged representations in **List-I** with their characterizations in **List-II** :

	List-I		List-II
(a)	Initial	(i)	Processes are improved quantitatively and continually.
(b)	Repeatable	(ii)	The plan for a project comes from a template for plans.
(c)	Defined	(iii)	The plan uses processes that can be measured quantitatively.
(d)	Managed	(iv)	There may not exist a plan or it may be abandoned.
(e)	Optimizing	(v)	There's a plan and people stick to it.

Code :

A. (a)-(iv); (b)-(v); (c)-(i); (d)-(iii); (e)-(ii)
B. (a)-(i); (b)-(ii); (c)-(iv); (d)-(v); (e)-(iii)
C. (a)-(v); (b)-(iv); (c)-(ii); (d)-(iii); (e)-(i)
D. (a)-(iv); (b)-(v); (c)-(ii); (d)-(iii); (e)-(i)

ugcnetcse-july2018-paper2 is&software-engineering cmmi software-development-life-cycle-models

Answer key

28.7 Cocomo Model (3)

28.7.1 Cocomo Model: UGC NET CSE | December 2008 | Part 2 | Question: 45

$COCOMO$ model is used for:

A. product quality estimation
B. product complexity estimation
C. product cost estimation
D. all of the above

ugcnetcse-dec2008-paper2 is&software-engineering cocomo-model

Answer key

28.7.2 Cocomo Model: UGC NET CSE | December 2010 | Part 2 | Question: 44

The COCOMO model was introduced in the book title "Software Engineering Economics" authored by

A. Abraham Silberschatz
B. Barry Boehm
C. C.J. Date
D. D.E. Knuth

ugcnetcse-dec2010-paper2 is&software-engineering cocomo-model

Answer key

28.7.3 Cocomo Model: UGC NET CSE | January 2017 | Part 3 | Question: 48

A software company needs to develop a project that is estimated as 1000 function points and is planning to use JAVA as the programming language whose approximate lines of code per function point is accepted as 50. Considering $a = 1.4$ as multiplicative factor, $b = 1.0$ as exponention factor for the basic COCOMO effort equation and $c = 3.0$ as multiplicative factor, $d = 0.33$ as exponention factor for the basic COCOMO duration equation, approximately how long does project take to complete?

A. 11.2 months
B. 12.2 months
C. 13.2 months
D. 10.2 months

ugcnetcse-jan2017-paper3 is&software-engineering cocomo-model

Answer key

28.8 Coding and Testing (1)

28.8.1 Coding and Testing: UGC NET CSE | July 2016 | Part 3 | Question: 46

Software safety is quality assurance activity that focuses on hazards that

A. affects the reliability of a software component
B. may cause an entire system to fail
C. may result from user input errors
D. prevent profitable marketing of the final product

ugcnetcse-july2016-paper3 is&software-engineering coding-and-testing

Answer key

28.9 Cohesion (2)

28.9.1 Cohesion: UGC NET CSE | December 2014 | Part 3 | Question: 48

Temporal cohesion means

A. Coincidental cohesion
B. Cohesion between temporary variables
C. Cohesion between local variables
D. Cohesion with respect to time

ugcnetcse-dec2014-paper3 is&software-engineering cohesion

Answer key

28.9.2 Cohesion: UGC NET CSE | Junet 2015 | Part 2 | Question: 44

Cohesion is an extension of:

A. Abstraction concept
B. Refinment concept
C. Information hiding concept
D. Modularity

ugcnetcse-june2015-paper2 is&software-engineering cohesion

Answer key

28.10 Cost Estimation Model (2)

28.10.1 Cost Estimation Model: UGC NET CSE | December 2009 | Part 2 | Question: 43

Recorded software attributes can be used in the following endeavours :
(i) Cost and schedule estimates.
(ii) Software product reliability predictions.
(iii) Managing the development process.
(iv) No where

Codes :
(A) (i) (ii) (iv)
(B) (ii) (iii) (iv)
(C) (i) (ii) (iii)
(D) (i) (ii) (iii) (iv)

ugcnetcse-dec2009-paper2 is&software-engineering cost-estimation-model

Answer key

28.10.2 Cost Estimation Model: UGC NET CSE | December 2013 | Part 3 | Question: 5

Function points can be calculated by

A. UFP * CAF
B. UFP * FAC
C. UFP * Cost
D. UFP * Productivity

ugcnetcse-dec2013-paper3 is&software-engineering cost-estimation-model

Answer key

28.11 Coupling (3)

28.11.1 Coupling: UGC NET CSE | December 2013 | Part 3 | Question: 6

Match the following :

	List – I		List – II
a.	Data Coupling	i.	Module A and Module B have shared data
b.	Stamp Coupling	ii.	Dependency between modules is based on the fact they communicate by only passing of data.
c.	Common Coupling	iii.	When complete data structure is passed from one module to another
d.	Content Coupling	iv.	When the control is passed from one module to the middle of another

Codes :

A. a-iii, b-ii, c-i, d-iv
B. a-ii, b-iii, c-i, d-iv
C. a-ii, b-iii, c-iv, d-i
D. a-iii, b-ii, c-iv, d-i

ugcnetcse-dec2013-paper3 is&software-engineering coupling

Answer key ☟

28.11.2 Coupling: UGC NET CSE | July 2018 | Part 2 | Question: 16

Coupling is a measure of the strength of the interconnections between software modules. Which of the following are correct statements with respect to module coupling?

P: Common coupling occurs when one module controls the flow of another module by passing it information on what to do

Q: In data coupling, the complete data structure is passed from one module to another through parameters.

R: Stamp coupling occurs when modules share a composite data structure and use only parts of it.

Code:

A. P and Q only
B. P and R only
C. Q and R only
D. All of P, Q and R

ugcnetcse-july2018-paper2 is&software-engineering coupling

Answer key ☟

28.11.3 Coupling: UGC NET CSE | September 2013 | Part 3 | Question: 46

In _____, modules A and B make use of a common data type, but perhaps perform different operations on it.

A. Data coupling
B. Stamp coupling
C. Control coupling
D. Content coupling

ugcnetcse-sep2013-paper3 is&software-engineering coupling

Answer key ☟

28.12 Coupling Cohesion (1)

28.12.1 Coupling Cohesion: UGC NET CSE | June 2019 | Part 2 | Question: 51

Which of the following statements is/are true?

P: In software engineering, defects that are discovered earlier are more expensive to fix.

Q: A software design is said to be a good design, if the components are strongly cohesive and weakly coupled.

Select the correct answer from the options given below:

A. P only
B. Q only
C. P and Q
D. Neither P nor Q

ugcnetcse-june2019-paper2 coupling-cohesion defect

Answer key ☟

28.13 Cyclomatic Complexity (2)

28.13.1 Cyclomatic Complexity: UGC NET CSE | December 2014 | Part 2 | Question: 45

Which one of the following is used to compute cyclomatic complexity ?

A. The number of regions -1
B. $E-N+1$, where E is the number of flow graph edges and N is the number of flow graph nodes.
C. $P-1$, where P is the number of predicate nodes in the flow graph G.
D. $P+1$, where P is the number of predicate nodes in the flow graph G.

ugcnetcse-dec2014-paper2 is&software-engineering cyclomatic-complexity

Answer key ☟

28.13.2 Cyclomatic Complexity: UGC NET CSE | July 2016 | Part 2 | Question: 44

The cyclomatic complexity of a flow graph V(G), in terms of predicate notes is

A. P+1
B. P-1
C. P-2
D. P+2

Where P is the number of predicate nodes in flow graph V(G)

ugcnetcse-july2016-paper2 cyclomatic-complexity

Answer key ☟

28.14 Data Mining (2)

28.14.1 Data Mining: UGC NET CSE | December 2012 | Part 3 | Question: 5

______ is process of extracting previously non known valid and actionable information from large data to make crucial business and strategic decisions.

A. Data Management
B. Data base
C. Data Mining
D. Meta Data

ugcnetcse-dec2012-paper3 data-mining

Answer key ☟

28.14.2 Data Mining: UGC NET CSE | June 2012 | Part 3 | Question: 12

Which of the following can be used for clustering the data?

A. Single layer perception
B. Multilayer perception
C. Self organizing map
D. Radial basis function

ugcnetcse-june2012-paper3 data-mining

Answer key ☟

28.15 Digital Marketing (1)

28.15.1 Digital Marketing: UGC NET CSE | December 2012 | Part 3 | Question: 1

Eco system is a framework for

A. Building a Computer System
B. Building Internet Market
C. Building Offline Market
D. Building Market

ugcnetcse-dec2012-paper3 e-technologies digital-marketing

Answer key ☟

28.16 Fan In (1)

28.16.1 Fan In: UGC NET CSE | December 2014 | Part 3 | Question: 47

$'FANIN'$ of a component A is defined as

A. Count of the number of components that can call, or pass control, to a component A
B. Number of components related to component A
C. Number of components dependent on component A
D. None of the above

ugcnetcse-dec2014-paper3 is&software-engineering fan-in

Answer key ☟

28.17 Flow Chart (1)

28.17.1 Flow Chart: UGC NET CSE | December 2010 | Part 2 | Question: 45

The Warnier diagram enables analyst

A. To represent information hierarchy in a compact manner
B. To further identify requirement
C. To estimate the total cost involved
D. None of the above

ugcnetcse-dec2010-paper2 is&software-engineering flow-chart

Answer key ☟

28.18 Function Point Metric (2)

28.18.1 Function Point Metric: UGC NET CSE | December 2014 | Part 3 | Question: 44

To compute function points (FP), the following relationship is used FP = Count - total $\times(0.65 + 0.01 \times \sum(F_i))$ where $F_i(i = 1 \text{ to } n)$ are value adjustment factors (VAF) based on n questions. The value of n is

A. 12
B. 14
C. 16
D. 18

ugcnetcse-dec2014-paper3 is&software-engineering function-point-metric

Answer key ☟

28.18.2 Function Point Metric: UGC NET CSE | June 2014 | Part 2 | Question: 14

In function point analysis, the number of complexity adjustment factors is

A. 10
B. 12
C. 14
D. 20

ugcnetcse-june2014-paper2 is&software-engineering function-point-metric

Answer key ☟

28.19 Git (1)

28.19.1 Git: UGC NET CSE | June 2019 | Part 2 | Question: 55

Which of the following terms best describes Git?

A. Issue Tracking System
B. Integrated Development Environment
C. Distributed Version Control System
D. Web-based Repository Hosting Service

ugcnetcse-june2019-paper2 git

Answer key ☟

28.20 Information System (1)

28.20.1 Information System: UGC NET CSE | December 2014 | Part 3 | Question: 60

A computer based information system is needed :

I. As it is difficult for administrative staff to process data.

II. Due to rapid growth of information and communication technology.

III. Due to growing size of organizations which need to process large volume of data.

IV. As timely and accurate decisions are to be taken.

Which of the above statement(s) is/are true ?

A. I and II B. III and IV C. II and III D. II and IV

ugcnetcse-dec2014-paper3 information-system

Answer key

28.21 Is&software Engineering (62)

28.21.1 Is&software Engineering: UGC NET CSE OCTOBER 2022

Which of the following are the products of grounded theory?

A. Experimental framework

B. Concepts

C. Hypotheses

D. Attributes of a category

E. Theoretical dilution

is&software-engineering

Answer key

28.21.2 Is&software Engineering: UGC NET CSE | August 2016 | Part 3 | Question: 47

Which of the following are external qualities of a software product ?

A. Maintainability, reusability, portability, efficiency, correctness.
B. Correctness, reliability, robustness, efficiency, usability.
C. Portability, interoperability, maintainability, reusability.
D. Robustness, efficiency, reliability, maintainability, reusability.

ugcnetcse-aug2016-paper3 is&software-engineering

Answer key

28.21.3 Is&software Engineering: UGC NET CSE | August 2016 | Part 3 | Question: 48

Which of the following is/are CORRECT statement(s) about version and release ?

I. A version is an instance of a system, which is functionally identical but nonfunctionally distinct from other instances of a system.

II. A version is an instance of a system, which is functionally distinct in some way from other system instances.

III. A release is an instance of a system, which is distributed to users outside of the development team.

IV. A release is an instance of a system, which is functionally identical but nonfunctionally distinct from other instances of a system.

A. I and III B. II and IV C. I and IV D. II and III

ugcnetcse-aug2016-paper3 is&software-engineering

Answer key

28.21.4 Is&software Engineering: UGC NET CSE | December 2011 | Part 2 | Question: 14

An SRS

A. Establishes the basis for agreement between client and the supplier.
B. Provides a reference for validation of the final product.
C. Is a prerequisite to high quality software.
D. All of the above.

ugcnetcse-dec2011-paper2 is&software-engineering

Answer key

28.21.5 Is&software Engineering: UGC NET CSE | December 2011 | Part 2 | Question: 17

Design recovery from source code is done during

A. reverse engineering
B. re-engineering
C. reuse
D. all of the above

ugcnetcse-dec2011-paper2 is&software-engineering

Answer key

28.21.6 Is&software Engineering: UGC NET CSE | December 2012 | Part 2 | Question: 19

The maturity levels used to measure a process are

A. Initial, Repeatable, Defined, Managed, Optimized
B. Primary, Secondary, Defined, Managed, Optimized
C. Initial, Stating, Defined, Managed, Optimized
D. None of the above

ugcnetcse-dec2012-paper2 non-gatecse is&software-engineering

Answer key

28.21.7 Is&software Engineering: UGC NET CSE | December 2012 | Part 2 | Question: 6

Component level design is concerned with

A. Flow oriented analysis
B. Class based analysis
C. Both of the above
D. None of the above

ugcnetcse-dec2012-paper2 non-gatecse is&software-engineering

Answer key

28.21.8 Is&software Engineering: UGC NET CSE | December 2013 | Part 2 | Question: 10

Testing of software with actual data and in actual environment is called

A. Alpha testing
B. Beta testing
C. Regression testing
D. None of the above

ugcnetcse-dec2013-paper2 is&software-engineering

Answer key

28.21.9 Is&software Engineering: UGC NET CSE | December 2013 | Part 2 | Question: 7

The relationship of data elements in a module is called

A. Coupling
B. Modularity
C. Cohesion
D. Granularity

ugcnetcse-dec2013-paper2 is&software-engineering

Answer key

28.21.10 Is&software Engineering: UGC NET CSE | December 2013 | Part 2 | Question: 8

Software Configuration Management is the discipline for systematically controlling

A. the changes due to the evolution of work products as the project proceeds
B. the changes due to defects (bugs) being found and then fixed
C. the changes due to requirement changes
D. all of the above

ugcnetcse-dec2013-paper2 is&software-engineering

Answer key

28.21.11 Is&software Engineering: UGC NET CSE | December 2013 | Part 2 | Question: 9

Which one of the following is not a step of requirement engineering?

A. Requirement elicitation
B. Requirement analysis
C. Requirement design
D. Requirement documentation

ugcnetcse-dec2013-paper2 is&software-engineering

Answer key

28.21.12 Is&software Engineering: UGC NET CSE | December 2013 | Part 3 | Question: 9

Which of the following is not a software myth?

A. Once we write the program and get it to work, our job is done
B. Projects requirements continually change, but change can be easily accommodated because software is flexible
C. If we get behind schedule, we can add more programmers and catch up
D. If an organization does not understand how to control software projects internally, it will invariably struggle when it outsources software projects.

ugcnetcse-dec2013-paper3 is&software-engineering

Answer key

28.21.13 Is&software Engineering: UGC NET CSE | December 2015 | Part 3 | Question: 25

Which of the following non-functional quality attributes is not highly affected by the architecture of the software?

A. Performance
B. Reliablity
C. Usability
D. Portability

ugcnetcse-dec2015-paper3 is&software-engineering

Answer key

28.21.14 Is&software Engineering: UGC NET CSE | December 2015 | Part 3 | Question: 72

Which one of the following statements is incorrect?

A. Pareto analysis is a statistical method used for analyzing causes, and is one of the primary tools for quality management.
B. Reliability of a software specifies the probability of failure-free operation of that software for a given time duration.
C. The reliability of a system can also be specified as the Mean Time To Failure (MTTF).
D. In white-box testing, the test cases are decided from the specifications or the requirements.

ugcnetcse-dec2015-paper3 is&software-engineering

Answer key

28.21.15 Is&software Engineering: UGC NET CSE | December 2015 | Part 3 | Question: 74

Which one of the following statements, related to the requirements phase in Software Engineering, is incorrect?

A. "Requirement validation" is one of the activities in the requirements phase.
B. "Prototyping" is one of the methods for requirement analysis.
C. "Modelling-oriented approach" is one of the methods for specifying the functional specifications.
D. "Function points" is one of the most commonly used size metric for requirements.

ugcnetcse-dec2015-paper3 is&software-engineering

Answer key

28.21.16 Is&software Engineering: UGC NET CSE | January 2017 | Part 2 | Question: 41

Software Engineering is an engineering discipline that is concerned with:

A. how computer systems work.
B. theories and methods that underlie computers and software systems.
C. all aspects of software production
D. all aspects of computer-based systems development, including hardware, software and process engineering.

ugcnetjan2017ii is&software-engineering

Answer key

28.21.17 Is&software Engineering: UGC NET CSE | January 2017 | Part 2 | Question: 42

Which of the following is not one of three software product aspects addressed by McCall's software quality factors ?

A. Ability to undergo change
B. Adaptiability to new environments
C. Operational characteristics
D. Production costs and scheduling

ugcnetjan2017ii is&software-engineering

Answer key

28.21.18 Is&software Engineering: UGC NET CSE | January 2017 | Part 2 | Question: 43

Which of the following statement(s) is/are true with respect to software architecture?

$S1$: Coupling is a measure of how well the things grouped together in a module belong together logically.

$S2$: Cohesion is a measure of the degree of interaction between software modules.

$S3$: If coupling is low and cohesion is high then it is easier to change one module without affecting others.

A. Only $S1$ and $S2$
B. Only $S3$
C. All of $S1$, $S2$ and $S3$
D. Only $S1$

ugcnetjan2017ii is&software-engineering

Answer key

28.21.19 Is&software Engineering: UGC NET CSE | January 2017 | Part 2 | Question: 44

The prototyping model of software development is:

A. a reasonable approach when requirements are well-defined.
B. a useful approach when a customer cannot define requirements clearly.
C. the best approach to use for projects with large development teams.
D. a risky model that rarely produces a meaningful product.

ugcnetjan2017ii is&software-engineering

Answer key

28.21.20 Is&software Engineering: UGC NET CSE | January 2017 | Part 3 | Question: 45

Match the terms related to Software Configuration Management (SCM) in List-I with the descriptions in List-II.

	List-I		List-II
I.	Version	A.	An instance of a system that is distributed to customers
II.	Release	B.	An instance of a system which is functionally identical to other instances but designed for different hardware/software instances.
III.	Variant	C.	An instance of a system that differs, in some way, from other instances

Codes:

A. I-B; II-C; III-C
B. I-C; II-A; III-B
C. I-C; II-B; III-A
D. I-B; II-A; III-C

ugcnetcse-jan2017-paper3 is&software-engineering

Answer key

28.21.21 Is&software Engineering: UGC NET CSE | January 2017 | Part 3 | Question: 46

A software project was estimated at 352 Function Points (FP). A four person team will be assigned to this project consisting of an architect, two programmers, and a tester The salary of the architect is ₹80,000 per month, the programmer ₹60,000 per month and the tester ₹50,000 per month. The average productivity for the team is 8 FP per person month. Which of the following represents the projected cost of the project?

A. ₹28,16,000
B. ₹20,90,000
C. ₹26,95,000
D. ₹27,50,000

ugcnetcse-jan2017-paper3 is&software-engineering

Answer key

28.21.22 Is&software Engineering: UGC NET CSE | January 2017 | Part 3 | Question: 47

Complete each of the following sentences in List-I on the left hand side by filling in the word or phrase from the List –II on the right hand side that best completes the sentence :

	List-I		List-II
I.	Determining whether you have built the right system is called ______	A.	Software testing
II.	Determining whether you have built the system right is called ______	B.	Software verification
III.	______ is the process of demostrating	C.	Software debuggingthe existence of defects
IV.	_______ is the process of discovering the cause of a defect and fixing it	D.	Software validation

Codes :

A. I-B, II-D, III-A, IV-C
B. I-B, II-D, III-C, IV-A
C. I-D, II-B, III-C, IV-A
D. I-D, II-B, III-A, IV-C

ugcnetcse-jan2017-paper3 is&software-engineering

Answer key

28.21.23 Is&software Engineering: UGC NET CSE | July 2016 | Part 2 | Question: 41

If S_1 is total number of modules defined in the program architecture, S_3 is the number of modules whose correct function depends on prior processing then the number of modules not dependent on prior processing is:

A. $1+\frac{S_3}{S_1}$
B. $1-\frac{S_3}{S_1}$
C. $1+\frac{S_1}{S_3}$
D. $1-\frac{S_1}{S_3}$

ugcnetcse-july2016-paper2 is&software-engineering

Answer key

28.21.24 Is&software Engineering: UGC NET CSE | July 2016 | Part 3 | Question: 44

Match the software maintenance activities in List-I to its meaning in List-II.

	List-I		List-II
a.	Corrective	i.	Concerning with performing activities to reduce the software complexity thereby improving program understandability and increasing software maintainability
b.	Adaptive	ii.	Concerned with fixing errors that are observed when the software is in use
c.	Perfective	iii.	Concerned with the change in the software that takes place to make the software adaptable to new environment (both hardware and software).
d.	Preventive	iv.	Concerned with the change in the software that takes place to make the software adaptable to changing user requirements

Codes :

A. i-b,ii-d,iii-c,iv-a
B. i-b,ii-c,iii-d,iv-a
C. i-c,ii-b,iii-d,iv-a
D. i-a,ii-d,iii-b,iv-c

ugcnetcse-july2016-paper3 is&software-engineering

Answer key

28.21.25 Is&software Engineering: UGC NET CSE | July 2018 | Part 2 | Question: 12

Match the following in Software Engineering :

	List-I		List-II
(a)	Product Complexity	(i)	Software Requirement Definition
(b)	Structured System Analysis	(ii)	Software Design
(c)	Coupling and Cohesion	(iii)	Validation Technique
(d)	Symbolic Execution	(iv)	Software Cost Estimation

Code :

A. (a)-(ii); (b)-(iii); (c)-(iv); (d)-(i)
B. (a)-(iii); (b)-(i); (c)-(iv); (d)-(ii)
C. (a)-(iv); (b)-(i); (c)-(ii); (d)-(iii)
D. (a)-(iii); (b)-(iv); (c)-(i); (d)-(ii)

ugcnetcse-july2018-paper2 is&software-engineering match-the-following

Answer key

28.21.26 Is&software Engineering: UGC NET CSE | July 2018 | Part 2 | Question: 13

Which one of the following is not typically provided by Source Code Management Software?

A. Synchronization
B. Versioning and Revision history
C. Syntax highlighting
D. Project forking

ugcnetcse-july2018-paper2 is&software-engineering

Answer key

28.21.27 Is&software Engineering: UGC NET CSE | July 2018 | Part 2 | Question: 14

A software system crashed 20 times in the year 2017 and for each crash, it took 2 minutes to restart. Approximately, what was the software availability in that year?

A. 96.9924% B. 97.9924% C. 98.9924% D. 99.9924%

ugcnetcse-july2018-paper2 is&software-engineering

Answer key

28.21.28 Is&software Engineering: UGC NET CSE | July 2018 | Part 2 | Question: 18

Reasons to re-engineer a software include:

P: Allow legacy software to quickly adapt to the changing requirements

Q: Upgrade to newer technologies/platforms/paradigm (for example, object-oriented)

R: Improve software maintainability

S: Allow change in the functionality and architecture of the software

Code:

A. P, R and S only B. P and R only C. P, Q and S only D. P, Q and R only

ugcnetcse-july2018-paper2 is&software-engineering

Answer key

28.21.29 Is&software Engineering: UGC NET CSE | July 2018 | Part 2 | Question: 19

Which of the following is not a key strategy followed by the clean room approach to software development?

A. Formal specification
B. Dynamic verification
C. Incremental development
D. Statistical testing of the system

ugcnetcse-july2018-paper2 is&software-engineering

Answer key

28.21.30 Is&software Engineering: UGC NET CSE | July 2018 | Part 2 | Question: 20

Which of the following statements is/are True?

P: Refactoring is the process of changing a software system in such a way that it does not alter the external behavior of the code yet improves the internal architecture

Q: An example of refactoring is adding new features to satisfy a customer requirement discovered after a project is shipped

Code:

A. P only B. Q only C. Both P and Q D. Neither P nor Q

ugcnetcse-july2018-paper2 is&software-engineering

Answer key

28.21.31 Is&software Engineering: UGC NET CSE | June 2005 | Part 2 | Question: 42

Which one of the following is not a software process model?

A. Linear sequential model
B. Prototyping model
C. The spiral model
D. $COCOMO$ model

ugcnetcse-june2005-paper2 is&software-engineering

Answer key

28.21.32 Is&software Engineering: UGC NET CSE | June 2007 | Part 2 | Question: 43

Match the following:

(a) Unit test	(i) Requirements
(b) System test	(ii) Design
(c) Validation test	(iii) Code
(d) Integration test	(iv) System Engineering

Which of the following is true:

A. a-ii, b-iii, c-iv, d-i
B. a-i, b-ii, c-iv, d-iii
C. a-iii, b-iv, c-i, d-ii
D. None of the above

ugcnetcse-june2007-paper2 match-the-following is&software-engineering non-gatecse

Answer key

28.21.33 Is&software Engineering: UGC NET CSE | June 2010 | Part 2 | Question: 41

Software engineering primarily aims on

A. Reliable software
B. Cost effective software
C. Reliable and cost effective software
D. None of the above

ugcnetcse-june2010-paper2 is&software-engineering

Answer key

28.21.34 Is&software Engineering: UGC NET CSE | June 2012 | Part 2 | Question: 20

Main aim of software engineering is to produce

A. program
B. software
C. within budget
D. software within budget in the given schedule

ugcnetcse-june2012-paper2 non-gatecse is&software-engineering

Answer key

28.21.35 Is&software Engineering: UGC NET CSE | June 2012 | Part 2 | Question: 21

Key process areas of CMM level 4 are also classified by a process which is

A. CMM level 2
B. CMM level 3
C. CMM level 5
D. All of the above

ugcnetcse-june2012-paper2 non-gatecse is&software-engineering

Answer key

28.21.36 Is&software Engineering: UGC NET CSE | June 2012 | Part 2 | Question: 22

Validation means

A. are we building the product right
B. are we building the right product
C. verification of fields
D. None of the above

ugcnetcse-june2012-paper2 is&software-engineering non-gatecse

Answer key

28.21.37 Is&software Engineering: UGC NET CSE | June 2012 | Part 2 | Question: 23

If a process is under statistical control, then it is

A. Maintainable
B. Measurable
C. Predictable
D. Verifiable

ugcnetcse-june2012-paper2 non-gatecse is&software-engineering

Answer key

28.21.38 Is&software Engineering: UGC NET CSE | June 2012 | Part 2 | Question: 24

In a function oriented design, we

A. minimize cohesion and maximize coupling
B. maximize cohesion and minimize coupling
C. maximize cohesion and maximize coupling
D. minimize cohesion and minimize coupling

ugcnetcse-june2012-paper2 non-gatecse is&software-engineering

Answer key

28.21.39 Is&software Engineering: UGC NET CSE | June 2012 | Part 2 | Question: 25

Which of the following metric does not depend on the programming language used?

A. Line of code
B. Function count
C. Member of token
D. All of the above

non-gatecse is&software-engineering ugcnetcse-june2012-paper2

Answer key

28.21.40 Is&software Engineering: UGC NET CSE | June 2012 | Part 2 | Question: 43

Reliability of software is directly dependent on

A. quality of the design
B. number of errors present
C. software engineers experience
D. user requirement

non-gatecse is&software-engineering ugcnetcse-june2012-paper2

Answer key

28.21.41 Is&software Engineering: UGC NET CSE | June 2012 | Part 3 | Question: 37

Are we building the right product? – This statement refers to

A. Verification
B. Validation
C. Testing
D. Software quality assurance

ugcnetcse-june2012-paper3 is&software-engineering

Answer key

28.21.42 Is&software Engineering: UGC NET CSE | June 2013 | Part 2 | Question: 3

While estimating the cost of software, Lines Of Code (LOC) and Function Points (FP) are used to measure which one of the following?

A. Length of code
B. Size of software
C. Functionality of software
D. None of the above

is&software-engineering ugcnetcse-june2013-paper2

Answer key

28.21.43 Is&software Engineering: UGC NET CSE | June 2014 | Part 2 | Question: 11

KPA in CMM stands for

A. Key Process Area
B. Key Product Area
C. Key Principal Area
D. Key Performance Area

ugcnetcse-june2014-paper2 is&software-engineering

Answer key

28.21.44 Is&software Engineering: UGC NET CSE | June 2014 | Part 2 | Question: 13

_________ of a system is the structure or structures of the system which comprise software elements, the externally visible properties of these elements and the relationship amongst them.

A. Software construction
B. Software evolution
C. Software architecture
D. Software reuse

ugcnetcse-june2014-paper2 is&software-engineering

Answer key

28.21.45 Is&software Engineering: UGC NET CSE | June 2014 | Part 2 | Question: 28

A software agent is defined as

I. A software developed for accomplishing a given task.
II. A computer program which is capable of acting on behalf of the user in order to accomplish a given computational task.
III. An open source software for accomplishing a given task.

A. I
B. II
C. III
D. All of the above

ugcnetcse-june2014-paper2 is&software-engineering

Answer key

28.21.46 Is&software Engineering: UGC NET CSE | June 2014 | Part 3 | Question: 19

Match the following :

	List – I		List – II
a.	Correctness	i.	The extent to which a software tolerates the unexpected problems.
b.	Accuracy	ii.	The extent to which a software meets its specification
c.	Robustness	iii.	The extent to which a software has the specified functions
d.	Completeness	iv.	Meeting specifications with precision

Codes :

A. a-ii, b-iv, c-i, d-iii
B. a-i, b-ii, c-iii, d-iv
C. a-ii, b-i, c-iv, d-iii
D. a-iv, b-ii, c-i, d-iii

ugcnetjune2014iii is&software-engineering

Answer key

28.21.47 Is&software Engineering: UGC NET CSE | June 2014 | Part 3 | Question: 20

Which one of the following is not a definition of error ?

A. It refers to the discrepancy between a computed, observed or measured value and the true, specified or theoretically correct value.
B. It refers to the actual output of a software and the correct output.
C. It refers to a condition that causes a system to fail.
D. It refers to human action that results in software containing a defect or fault.

ugcnetjune2014iii is&software-engineering

Answer key

28.21.48 Is&software Engineering: UGC NET CSE | Junet 2015 | Part 3 | Question: 44

Which design matric is used to measure the compactness of the program in terms of lines of code?

A. Consistency
B. Conciseness
C. Efficiency
D. Accuracy

is&software-engineering non-gatecse ugcnetcse-june2015-paper3

Answer key

28.21.49 Is&software Engineering: UGC NET CSE | Junet 2015 | Part 3 | Question: 45

Requirements prioritization and negotiation belongs to

A. Requirements validation
B. Requirements elicitation

C. Feasibility study

D. Requirements reviews

is&software-engineering non-gatecse ugcnetcse-june2015-paper3

Answer key

28.21.50 Is&software Engineering: UGC NET CSE | Junet 2015 | Part 3 | Question: 60

Match the following :

	List-I		List-II
(a)	Joint Application Design (JAD)	(i)	Delivers functionality in rapid iteration measured in weeks and needs frequent communication, development, testing and delivery
(b)	Computer Aided Software Engineering	(ii)	Reusable applications generally with one specific function. It is closely linked with idea of web services and service oriented architecture
(c)	Agile development	(iii)	Tools to automate many tasks of SDLC
(d)	Component Based Technology	(iv)	A group based tool for collecting user requirements and creating system design. Mostly used in analysis and design stages of SDLC

Codes :

A. (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

B. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

C. (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

D. (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

ugcnetcse-june2015-paper3 is&software-engineering

Answer key

28.21.51 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 19

Which of the following UML diagrams has a static view?

A. Collaboration diagram

B. Use-Case diagram

C. State chart diagram

D. Activity diagram

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key

28.21.52 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 20

Modifying the software by restructuring is called

A. Adaptive maintenance

B. Corrective maintenance

C. Perfective maintenance

D. Preventive maintenance

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key

28.21.53 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 21

A company has a choice of two languages L_1 and L_2 to develop a software for their client. Number of LOC required to develop an application in L_2 is thrice the LOC in language L_1. Also, software has to be maintained for next 10 years. Various parameters for two languages are given below to decide which language should be preferred for development.

PARAMETER	L_1	L_2
Man-year needed for development	LOC/1000	LOC/1000
Development cost	Rs. 70, 000	Rs. 90, 000
Cost of Maintenance per year	Rs.1, 00, 000	Rs.40, 000

Total cost of project include cost of development and maintenance. What is the LOC for L_1 for which cost of developing the software with both languages must be same?

A. 2000 B. 6000 C. 3000 D. 5000

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key

28.21.54 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 22

A Software project was estimated at 864 Function Points. A six person team will be assigned to project consisting of a requirement gathering person, one designer, two programmers and two testers. The salary of the designer is Rs. 70, 000 per month, requirement gatherer is Rs. 50, 000 per month, programmer is Rs. 60, 000 per month and a tester is Rs. 60, 000 per month. Average productivity for the team is 12 FP per person month. Which of the following represents the projected cost of the project

A. Rs. 33, 20, 000
B. Rs. 43, 20, 000
C. Rs. 33, 10, 000
D. Rs. 22, 10, 000

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key

28.21.55 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 45

Which of the following statements regarding XML is/are True?

i. XML is a set of tags designed to tell browsers how to display text and images in a web page
ii. XML defines a syntax for representing data, but the meaning of data varies from application to application
iii. <Letter>, <LETTER> and <letter> are three different tags in XML

Choose the correct answer from the options given below:

A. (i) and (ii) only
B. (i) and (iii) only
C. (ii) and (iii) only
D. (i) (ii) and (iii)

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key

28.21.56 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 49

Software reliability is described with respect to

i. Execution Time
ii. Calender Time
iii. Clock Time

Choose the correct answer from the options given below:

A. (i) and (ii) only
B. (ii) and (iii) only
C. (i), (ii) and (iii)
D. (i) and (iii) only

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key

28.21.57 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 50

To create an object-behavioral model, the analyst performs the following steps:

a. Evaluates all use-cases
b. Builds state transition diagram for the system
c. Reviews the object behaviour model to verify accuracy
d. Identifies events that do not derive the interaction

and consistency sequence

Choose the correct answer from the options given below:

A. (a), (b) and (c) only
B. (a), (b) and (d) only
C. (b), (c) and (d) only
D. (a), (c) and (d) only

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key

28.21.58 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 51

Which of the following is/are behavioral testing technique(s)?

a. Equivalence Partitioning
b. Graph-Based Treating Method
c. Boundary Value Analysis
d. Data flow Testing
e. Loop Testing

Choose the correct answer from the options given below:

A. (b) and (d) only
B. (a), (b) and (c) only
C. (d) and (e) only
D. (a), (c) and (e) only

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key

28.21.59 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 68

Match list I with List II

With reference to CMM developed by Software Engineering Institute (SEI)

	List I		List II
(A)	INITIAL	(I)	Process measurement
(B)	REPEATABLE	(II)	Process definition
(C)	DEFINED	(III)	Project management
(D)	MANAGED	(IV)	ADHOC

Choose the correct answer from the options given below:

A. $A-III, B-IV, C-II, D-I$
B. $A-IV, B-III, C-I, D-II$
C. $A-IV, B-III, C-II, D-I$
D. $A-III, B-IV, C-I, D-II$

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key

28.21.60 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 77

Arrange the following types of Cohesion from the best to worst type.

a. Logical Cohesion
b. Sequential Cohesion
c. Communication Cohesion
d. Temporal Cohesion
e. Procedural Cohesion

Choose the correct answer from the options given below:

A. $a \rightarrow d \rightarrow e \rightarrow c \rightarrow b$
B. $a \rightarrow e \rightarrow d \rightarrow c \rightarrow b$
C. $b \rightarrow e \rightarrow c \rightarrow d \rightarrow a$
D. $b \rightarrow c \rightarrow e \rightarrow d \rightarrow a$

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key

28.21.61 Is&software Engineering: UGC NET CSE | October 2020 | Part 2 | Question: 85

Statement I: Quality control involves the series of inspections, reviews and tests used throughout the software process, to ensure each work product meets the requirements placed upon it.

Statement II: Quality assurance consists of auditing and reporting functions of management

In the light of the above statements, choose the correct answer from the options given below

A. Both Statement I and Statement II are true
B. Both Statement I and Statement II are false
C. Statement I is correct but Statement II is false
D. Statement I is incorrect but Statement II is true

ugcnetcse-oct2020-paper2 non-gatecse is&software-engineering

Answer key ☟

28.21.62 Is&software Engineering: UGC NET CSE | September 2013 | Part 2 | Question: 28

The _____ of a program or computing system is the structure or structures of the system, which comprise software components, the externally visible properties of these components and the relationship among them.

A. E-R diagram
B. Data flow diagram
C. Software architecture
D. Software design

ugcnetsep2013ii is&software-engineering

Answer key ☟

28.22 Life Cycle Model (1)

28.22.1 Life Cycle Model: UGC NET CSE | Junet 2015 | Part 2 | Question: 43

Which process model is also called as classic life cycle model?

A. Waterfall model
B. RAD model
C. Prototyping model
D. Incremental model

ugcnetcse-june2015-paper2 is&software-engineering life-cycle-model

Answer key ☟

28.23 Mis (1)

28.23.1 Mis: UGC NET CSE | July 2016 | Part 3 | Question: 71

Which of the following is characteristic of MIS?

A. Provides guidance in identifying problems, finding and evaluating alternative solutions, and selecting or comparing alternatives
B. Draws on diverse yet predictable data resources to aggregate and summarize data
C. High volume, data capture focus
D. Has as its goal the efficiency of data movement and processing and interfacing different TPS

ugcnetcse-july2016-paper3 mis is&software-engineering

Answer key ☟

28.24 Model View Controller (1)

28.24.1 Model View Controller: UGC NET CSE | June 2019 | Part 2 | Question: 52

The M components in MVC are responsible for

A. user interface
B. security of the system
C. business logic and domain objects
D. translating between user interface actions/events and operations on the domain objects

ugcnetcse-june2019-paper2 model-view-controller

Answer key

28.25 Object Oriented Modelling (1)

28.25.1 Object Oriented Modelling: UGC NET CSE | December 2014 | Part 3 | Question: 41

What is true about UML stereotypes ?

A. Stereotype is used for extending the UML language.
B. Stereotyped class must be abstract
C. The stereotype indicates that the UML element cannot be changed
D. UML profiles can be stereotyped for backward compatibility

ugcnetcse-dec2014-paper3 object-oriented-modelling uml

Answer key

28.26 Optimization (1)

28.26.1 Optimization: UGC NET CSE | June 2012 | Part 3 | Question: 49

In any simplex table, if corresponding to any negative Δ j, all elements of the column are negative or zero, the solution under the test is

A. degenerate solution
B. unbounded solution
C. alternative solution
D. non-existing solution

ugcnetcse-june2012-paper3 optimization linear-programming

Answer key

28.27 Project Planning (1)

28.27.1 Project Planning: UGC NET CSE | Junet 2015 | Part 2 | Question: 45

Which one from the following is highly associated activity of project planning?

A. Keep track of the progress
B. Compare the actual and planned progress and costs
C. Identify the activities, milestones and deliverables produced by a project
D. Both B and C

ugcnetcse-june2015-paper2 is&software-engineering project-planning

Answer key

28.28 Project Tracking (1)

28.28.1 Project Tracking: UGC NET CSE | June 2019 | Part 2 | Question: 59

Which of the following are the primary objectives of risk monitoring in software project tracking?

P: To assess whether predicted risks do, in fact, occur

Q: To ensure that risk aversion steps defined for the risk are being properly applied

R: To collect information that can be used for future risk analysis

A. Only P and Q
B. Only P and R
C. Only Q and R
D. All of P, Q, R

ugcnetcse-june2019-paper2 project-tracking risk-monitoring

Answer key

28.29 Risk Management (3)

28.29.1 Risk Management: UGC NET CSE | December 2014 | Part 3 | Question: 45

Assume that the software team defines a project risk with 80% probability of occurrence of risk in the following manner: Only 70 percent of the software components scheduled for reuse will be integrated into the application and the remaining functionality will have to be custom developed. If 60 reusable components were planned with average component size as 100 LOC and software engineering cost for each LOC as $14, then the risk exposure would be

A. $25,200 B. $20,160 C. $17,640 D. $15,120

ugcnetcse-dec2014-paper3 is&software-engineering risk-management

Answer key

28.29.2 Risk Management: UGC NET CSE | June 2014 | Part 2 | Question: 12

Which one of the following is not a risk management technique for managing the risk due to unrealistic schedules and budgets ?

A. Detailed multi source cost and schedule estimation.
B. Design cost
C. Incremental development
D. Information hiding

ugcnetcse-june2014-paper2 is&software-engineering risk-management

Answer key

28.29.3 Risk Management: UGC NET CSE | September 2013 | Part 3 | Question: 50

Sixty reusable components were available for an application. If only 70% of these components can be used, rest 30% would have to be developed from scratch. If average component is 100 LOC and cost of each LOC is Rs. 14, what will be risk of exposure if risk probability is 80%?

A. Rs. 25, 200 B. Rs. 20, 160 C. Rs. 25, 160 D. Rs. 20, 400

ugcnetcse-sep2013-paper3 is&software-engineering risk-management

Answer key

28.30 Size Metrices (1)

28.30.1 Size Metrices: UGC NET CSE | Junet 2015 | Part 2 | Question: 41

Match the following:

a.	Size-oriented metrics	*i.*	uses number of external interfaces as one of the measurement parameters
b.	Function-oriented metrics	*ii.*	originally designed to be applied to business information systems
c.	Extended function point metrics	*iii.*	derived by normalizing quality and/or productivity measures by considering the size of the software
d.	Function point	*iv.*	uses algorithm characteristics as one of the measurement parameter

A. $a-iii, b-iv, c-i, d-ii$
B. $a-ii, b-i, c-iv, d-iii$
C. $a-iv, b-ii, c-iii, d-i$
D. $a-iii, b-i, c-iv, d-ii$

ugcnetcse-june2015-paper2 size-metrices is&software-engineering

Answer key

28.31 Software (1)

28.31.1 Software: UGC NET CSE | January 2017 | Part 2 | Question: 45

A software design pattern used to enhance the functionality of an object at run-time is:

A. Adapter B. Decorator C. Delegation D. Proxy

ugcnetjan2017ii software non-gatecse is&software-engineering

Answer key

28.32 Software Configuration (1)

28.32.1 Software Configuration: UGC NET CSE | December 2013 | Part 3 | Question: 7

A process which defines a series of tasks that have the following four primary objectives is known as

1. to identify all items that collectively define the software configuration
2. to manage changes to one or more of these items
3. to facilitate the construction of different versions of an application
4. to ensure that software quality is maintained as the configuration evolves over time

A. Software Quality Management Process
B. Software Configuration Management Process
C. Software Version Management Process
D. Software Change Management Process

ugcnetcse-dec2013-paper3 is&software-engineering software-configuration

Answer key

28.33 Software Cost Estimation (1)

28.33.1 Software Cost Estimation: UGC NET CSE | June 2014 | Part 3 | Question: 18

Consider a project with the following functional units :

Number of user inputs = 50

Number of user outputs = 40

Number of user enquiries = 35

Number of user files = 06

Number of external interfaces = 04

Assuming all complexity adjustment factors and weighing factors as average, the function points for the project will be

A. 135 B. 722 C. 675 D. 672

ugcnetjune2014iii is&software-engineering software-cost-estimation

Answer key

28.34 Software Design (18)

28.34.1 Software Design: Computer Science - UGC NET 2021 [Question ID = 2342]

The V components in MVC are responsible for :

1. User interface.
2. Security of the system.
3. Business logic and domain objects.
4. Translating between user interface actions / events and operations on the domain objects.

software-design

Answer key

28.34.2 Software Design: Computer Science - UGC NET 2021 [Question ID = 2344]

If every requirement stated in the Software Requirement Specification (SRS) has only one interpretation, then SRS is said to be :

1. correct
2. consistent
3. unambiguous
4. verifiable

software-design

Answer key

28.34.3 Software Design: Computer Science - UGC NET 2021 [Question ID = 2348]

Identify the correct order of the following five levels of Capability Maturity Model (from lower to higher) to measure the maturity of an organization's software process.

A. Defined
B. Optimizing
C. Initial
D. Managed
E. Repeatable

Choose the correct answer from the options given below

1. C, A, E, D, B
2. C, B, D, E, A
3. C,E, A, B, D
4. C, E, A, D, B

non-gatecse software-design

Answer key

28.34.4 Software Design: Computer Science - UGC NET 2021 [Question ID = 2350]

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : Software developers do not do exhaustive software testing in practice.
Reason R : Even for small inputs, exhaustive testing is too computationally intensive (e.g., takes too long) to run all the tests.

In light of the above statements, choose the correct answer from the options given below :

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

software-design software-testing non-gatecse

Answer key

28.34.5 Software Design: UGC NET CSE | August 2016 | Part 2 | Question: 43

The quick design of a software that is visible to end users leads to _____.

A. Iterative model
B. Prototype model
C. Spiral model
D. Waterfall model

ugcnetcse-aug2016-paper2 is&software-engineering software-design

Answer key

28.34.6 Software Design: UGC NET CSE | December 2011 | Part 2 | Question: 41

Software risk estimation involves following two tasks :

A. Risk magnitude and risk impact
B. Risk probability and risk impact
C. Risk maintenance and risk impact
D. Risk development and risk impact

ugcnetcse-dec2011-paper2 is&software-engineering software-design

Answer key

28.34.7 Software Design: UGC NET CSE | December 2014 | Part 2 | Question: 43

The software _______ of a program or a computing system is the structure or structures of the system, which comprise software components, the externally visible properties of those components, and the relationships among them.

A. Design
B. Architecture
C. Process
D. Requirement

ugcnetcse-dec2014-paper2 is&software-engineering software-design

Answer key

28.34.8 Software Design: UGC NET CSE | July 2016 | Part 3 | Question: 45

Match each application/software design concept in List-I to its definition in List-II.

	List-I		List-II
I.	Coupling	(a)	Easy to visually inspect the design of the software and understand its purpose
II.	Cohesion	(b)	Easy to add functionality to a software without having to redesign it
III.	Scalable	(c)	Focus of a code upon a single goal
IV.	Readable	(d)	Relaince of a code module upon other code modules

Codes :

A. I-b, II-a, III-d, IV-c
B. I-c, II-d, III-a, IV-b
C. I-d, II-c, III-b, IV-a
D. I-d, II-a, III-c, IV-b

ugcnetcse-july2016-paper3 is&software-engineering software-design

Answer key

28.34.9 Software Design: UGC NET CSE | July 2018 | Part 2 | Question: 17

A software design pattern often used to restrict access to an object is

A. adapter
B. decorator
C. delegation
D. proxy

ugcnetcse-july2018-paper2 is&software-engineering software-design

Answer key

28.34.10 Software Design: UGC NET CSE | June 2010 | Part 2 | Question: 42

Top-down design does not require

A. Step-wise refinement
B. Loop invariants
C. Flow charting
D. Modularity

ugcnetcse-june2010-paper2 is&software-engineering software-design

Answer key

28.34.11 Software Design: UGC NET CSE | June 2010 | Part 2 | Question: 44

Design phase will usually be

A. Top-down B. Bottom-up C. Random D. Centre fringing

ugcnetcse-june2010-paper2 is&software-engineering software-design

Answer key

28.34.12 Software Design: UGC NET CSE | June 2013 | Part 3 | Question: 1

The Software Maturity Index (SMI) is defined as $SMI = [M_f - (F_a + F_c + F_d)] / M_f$
where M_f= number of modules in current release
F_a =the number of module in the current release that have been added
F_c = the number of module in the current release that have been changed
F_d= the number of module in the current release that have been deleted
The product begins to stabilize when

A. SMI approaches 1
B. SMI approaches 0
C. SMI approaches -1
D. None of the above

ugcnetcse-june2013-paper3 is&software-engineering software-design non-gatecse

Answer key

28.34.13 Software Design: UGC NET CSE | June 2013 | Part 3 | Question: 10

Match the following :

a.	RAID 0	i.	Bit interleaved parity
b.	RAID 1	ii.	Non redundant stripping
c.	RAID 2	iii.	Mirrored disks
d.	RAID 3	iv.	Error correcting codes

Codes:

A. a-iv, b-i, c-ii, d-iii
B. a-iii, b-iv, c-i, d-ii
C. a-iii, b-i, c-iv, d-ii
D. a-iii, b-ii, c-iv, d-i

ugcnetcse-june2013-paper3 is&software-engineering software-design

Answer key

28.34.14 Software Design: UGC NET CSE | June 2013 | Part 3 | Question: 3

______ is a process model that removes defects before they can precipitate serious hazards.

A. Incremental model
B. Spiral model
C. Cleanroom software engineering
D. Agile model

ugcnetcse-june2013-paper3 is&software-engineering software-design

Answer key

28.34.15 Software Design: UGC NET CSE | June 2013 | Part 3 | Question: 5

The following three golden rules

- Place the user in control
- Reduce the user's memory load
- Make the interface consistent

are of

A. User satisfaction
B. Good interface design
C. Saving system's resources
D. None of these

ugcnetcse-june2013-paper3 is&software-engineering software-design

Answer key

28.34.16 Software Design: UGC NET CSE | Junet 2015 | Part 3 | Question: 42

Modulo design is used to maximize cohesion and minimize coupling. Which of the following is the key to implement this rule?

A. Inheritance
B. Polymorphism
C. Encapsulation
D. Abstraction

ugcnetcse-june2015-paper3 software-design is&software-engineering

Answer key

28.34.17 Software Design: UGC NET CSE | Junet 2015 | Part 3 | Question: 47

A Design Concept Refinement is a

A. Top-down approach
B. Complementary of abstraction concept
C. process of elaboration
D. All of the above

ugcnetcse-june2015-paper3 software-design is&software-engineering

Answer key

28.34.18 Software Design: UGC NET CSE | Junet 2015 | Part 3 | Question: 48

A software design is highly modular if

A. cohesion is functional and coupling is data type
B. cohesion is coincidental and coupling is data type
C. cohesion is sequential and coupling is content type
D. cohesion is functional and coupling is stamp type

ugcnetcse-june2015-paper3 is&software-engineering software-design

Answer key

28.35 Software Development (3)

28.35.1 Software Development: UGC NET CSE | December 2009 | Part 2 | Question: 43

Software Engineering is a discipline that integrates _________ for the development of computer software.

(A) Process

(B) Methods

(C) Tools

(D) All

ugcnetcse-dec2009-paper2 is&software-engineering software-development

Answer key

28.35.2 Software Development: UGC NET CSE | July 2018 | Part 2 | Question: 11

Assume the following regarding the development of a software system P:

- Estimated lines of codes P : 33, 480 LOC
- Average productivity for P : 620 LOC per person -month
- Number of software developers : 6
- Average salary of a software developer : INR 50, 000 per month

If E, D and C are the estimated development effort (in person-month), estimated development time (in months), and estimated development cost (in INR Lac) respectively, then (E, D, C) = ______

A. (48, 8, 24)   B. (54, 9, 27)   C. (60, 10, 30)   D. (42, 7, 21)

ugcnetcse-july2018-paper2   is&software-engineering   software-development

Answer key

28.35.3 Software Development: UGC NET CSE | September 2013 | Part 2 | Question: 27

In which one of the following continuous process improvement is done?

A. ISO9001   B. RMMM   C. CMM   D. None of the above

ugcnetsep2013ii   is&software-engineering   software-development

Answer key

28.36 Software Development Life Cycle Models (11)

28.36.1 Software Development Life Cycle Models: UGC NET CSE | August 2016 | Part 3 | Question: 44

Match each software lifecycle model in List – I to its description in List – II :

	List-I		List-II
I.	Code-and-Fix	a.	Assess risks at each step; do most critical action first
II.	Evolutionary Prototyping	b.	Build an initial small requirement specifications, code it, then "evolve" the specifications and code as needed
III.	Spiral	c.	Build initial requirement specification for several releases, then design-and-code in sequence
IV.	Stagged delivery	d.	Standard phases (requirements, design, code, test) in order
V.	Waterfall	e.	Write some code,debug it, repeat(i.e. ad-hoc)

Codes :

A. I-e, II-b, III-a, IV-c, V-d
B. I-e,II-c, III-a IV-b, V-d
C. I-d, II-a, III-b, IV-c, V-e
D. I-c, II-e, III-a, IV-b, V-d

ugcnetcse-aug2016-paper3   is&software-engineering   software-development-life-cycle-models

Answer key

28.36.2 Software Development Life Cycle Models: UGC NET CSE | December 2008 | Part 2 | Question: 44

Which level is called as "*defined*" in capability maturity model?

A. level 0
B. level 3
C. level 4
D. level 1

ugcnetcse-dec2008-paper2   is&software-engineering   software-development-life-cycle-models

Answer key

28.36.3 Software Development Life Cycle Models: UGC NET CSE | December 2010 | Part 2 | Question: 42

Which one of these are not software maintenance activity ?

A. Error correction
B. Adaptation
C. Implementation of Enhancement
D. Establishing scope

ugcnetcse-dec2010-paper2   is&software-engineering   software-development-life-cycle-models

Answer key

28.36.4 Software Development Life Cycle Models: UGC NET CSE | December 2010 | Part 2 | Question: 43

The system specification is the first deliverable in the computer system engineering process which does not include

A. Functional Description
B. Cost
C. Schedule
D. Technical Analysis

ugcnetcse-dec2010-paper2 is&software-engineering software-development-life-cycle-models

Answer key

28.36.5 Software Development Life Cycle Models: UGC NET CSE | December 2011 | Part 2 | Question: 13

For a data entry project for office staff who have never used computers before (user interface and user friendliness are extremely important), one will use

A. Spiral model
B. Component based model
C. Prototyping
D. Waterfall model

ugcnetcse-dec2011-paper2 is&software-engineering software-development-life-cycle-models

Answer key

28.36.6 Software Development Life Cycle Models: UGC NET CSE | December 2012 | Part 2 | Question: 24

RAD stands for

A. Rapid and Design
B. Rapid Aided Development
C. Rapid Application Development
D. Rapid Application Design

ugcnetcse-dec2012-paper2 is&software-engineering software-development-life-cycle-models

Answer key

28.36.7 Software Development Life Cycle Models: UGC NET CSE | December 2012 | Part 2 | Question: 48

_____ is an “umbrella” activity that is applied throughout the software engineering process.

A. Debugging
B. Testing
C. Designing
D. Software quality assurance

ugcnetcse-dec2012-paper2 is&software-engineering software-development-life-cycle-models

Answer key

28.36.8 Software Development Life Cycle Models: UGC NET CSE | December 2015 | Part 2 | Question: 47

Which of the following is not a software process model?

A. Prototyping
B. Iterative
C. Timeboxiing
D. Glassboxing

ugcnetcse-dec2015-paper2 is&software-engineering software-development-life-cycle-models

Answer key

28.36.9 Software Development Life Cycle Models: UGC NET CSE | July 2016 | Part 2 | Question: 42

The ____ model is preferred for software development when the requirements are not clear.

A. Rapid Application Development
B. Rational Unified Process
C. Evolutionary Model
D. Waterfall Model

ugcnetcse-july2016-paper2 is&software-engineering software-development-life-cycle-models

Answer key

28.36.10 Software Development Life Cycle Models: UGC NET CSE | June 2019 | Part 2 | Question: 57

Match List-I with List-II:

	List-I (Software Process Models)		List-II (Software Systems)
(a)	Waterfall model	(i)	e-business software that starts with only the basic functionalities and then moves on to more advanced features
(b)	Incremental development	(ii)	An inventory control system for super-market to be developed within three months
(c)	Prototyping	(iii)	A virtual reality system for simulating vehicle nevigation in a highway
(d)	RAD	(iv)	Automate the manual system for student record maintenance in a school

Choose the correct option from those given below:

A. a-ii; b-iv; c-i; d-iii
B. a-i; b-iii; c-iv; d-ii
C. a-iii; b-ii; c-iv; d-i
D. a-iv; b-i; c-iii; d-ii

ugcnetcse-june2019-paper2 software-development-life-cycle-models

Answer key

28.36.11 Software Development Life Cycle Models: UGC NET CSE | September 2013 | Part 2 | Question: 29

Working software is not available until late in the process in

A. Waterfall model
B. Prototyping model
C. Incremental model
D. Evolutionary Development model

ugcnetsep2013ii is&software-engineering software-development-life-cycle-models

Answer key

28.37 Software Development Models (1)

28.37.1 Software Development Models: UGC NET CSE | June 2010 | Part 2 | Question: 43

Which model is simplest model in Software Development ?

A. Waterfall model
B. Prototyping
C. Iterative
D. None of these

ugcnetcse-june2010-paper2 is&software-engineering software-development-models

Answer key

28.38 Software Intergrity (1)

28.38.1 Software Intergrity: UGC NET CSE | June 2019 | Part 2 | Question: 56

A Web application and its support environment has not been fully fortified against attack. Web engineers estimate that the likelihood of repelling an attack is only 30 percent. The application does not contain sensitive or controversial information, so the threat probability is 25 percent. What is the integrity of the web application?

A. 0.625
B. 0.725
C. 0.775
D. 0.825

ugcnetcse-june2019-paper2 software-intergrity

Answer key

28.39 Software Maintenance (2)

28.39.1 Software Maintenance: UGC NET CSE | Junet 2015 | Part 3 | Question: 46

Adaptive maintenance is a maintenance which _____

A. correct errors that were not discovered till testing phase
B. is carried out to port the existing software to a new environment
C. improves the system performance
D. both B and C

ugcnetcse-june2015-paper3 software-maintenance is&software-engineering

Answer key

28.39.2 Software Maintenance: UGC NET CSE | September 2013 | Part 3 | Question: 45

Improving processing efficiency or performance or restructuring of software to improve changeability is known as

A. Corrective maintenance
B. Perfective maintenance
C. Adaptive maintenance
D. Code maintenance

ugcnetcse-sep2013-paper3 is&software-engineering software-maintenance

Answer key

28.40 Software Metrics (3)

28.40.1 Software Metrics: UGC NET CSE | December 2014 | Part 2 | Question: 44

Which one of the following set of attributes should not be encompassed by effective software metrics ?

A. Simple and computable
B. Consistent and objective
C. Consistent in the use of units and dimensions
D. Programming language dependent

ugcnetcse-dec2014-paper2 is&software-engineering software-metrics

Answer key

28.40.2 Software Metrics: UGC NET CSE | July 2016 | Part 3 | Question: 48

The number of function points of a proposed system is calculated as 500. Suppose that the system is planned to be developed in Java and the LOC/FP ratio of Jave is 50. Estimate teh effort (E) required to complete the project using the effort formula of basic COCOMO given: E=a(KLOC)^b

Assume that the values of a and b are 2.5 and 1.0 respectively

A. 25 person months
B. 75 person months
C. 62.5 person months
D. 72.5 person months

ugcnetcse-july2016-paper3 is&software-engineering software-metrics

Answer key

28.40.3 Software Metrics: UGC NET CSE | June 2012 | Part 3 | Question: 33

Which one of the following statements is incorrect?

A. The number of regions corresponds to the cyclomatic complexity
B. Cyclomatic complexity for a flow graph G is $V(G) = N - E + 2$, where E is the number of edges and N is the number of nodes in flow graph.
C. Cyclomatic complexity for a flow graph G is $V(G) = E - N + 2$, where E is the number of edges and N is the number of nodes in flow graph.
D. Cyclomatic complexity for a flow graph G is $V(G) = P + 1$, where P is the number of predicate nodes contained in the flow graph G.

ugcnetcse-june2012-paper3 is&software-engineering software-metrics

Answer key

28.41 Software Process (1)

28.41.1 Software Process: UGC NET CSE | December 2014 | Part 2 | Question: 41

_________ are applied throughout the software process.

A. Framework activities
B. Umbrella activities
C. Planning activities
D. Construction activities

ugcnetcse-dec2014-paper2 is&software-engineering software-process

Answer key

28.42 Software Quality Assurance (2)

28.42.1 Software Quality Assurance: UGC NET CSE | August 2016 | Part 3 | Question: 46

The ISO quality assurance standard that applies to software Engineering is

A. ISO 9000 : 2004
B. ISO 9001 : 2000
C. ISO 9002 : 2001
D. ISO 9003 : 2004

ugcnetcse-aug2016-paper3 is&software-engineering software-quality-assurance

Answer key

28.42.2 Software Quality Assurance: UGC NET CSE | December 2012 | Part 3 | Question: 55

The factors that determine the quality of a software system are

A. correctness, reliability
B. efficiency, usability, maintainability
C. testability, portability, accuracy, error tolerances, expandability, access control, audit
D. All of the above

ugcnetcse-dec2012-paper3 is&software-engineering software-quality-assurance

Answer key

28.43 Software Quality Characteristics (1)

28.43.1 Software Quality Characteristics: UGC NET CSE | July 2016 | Part 2 | Question: 45

The extent to which a software tolerates the unexpected problems, is termed as

A. Accuracy
B. Reliability
C. Correctness
D. Robustness

ugcnetcse-july2016-paper2 software-quality-characteristics

Answer key

28.44 Software Reliability (11)

28.44.1 Software Reliability: Computer Science - UGC NET 2021 [Question ID = 2345]

A system has 99.99% uptime and has a mean-time-between-failure of 1 day. How fast does the system have to repair itself in order to reach this availability goal ?

1. 10 Seconds

2. 11 Seconds

3. 12 Seconds

4. 9 Seconds

software-design software-reliability software-testing

Answer key

28.44.2 Software Reliability: Computer Science - UGC NET 2021 [Question ID = 2346]

In the context of Software Configuration Management (SCM), what kind of files should be committed to your source control repository ?

A. Code files
B. Documentation files
C. Output files
D. Automatically generated files that are required for your system to be used

Choose the correct answer from the options given below:

1. A and B only
2. B and C only
3. C and D only
4. D and A only

software-design software-reliability non-gatecse

Answer key

28.44.3 Software Reliability: Computer Science - UGC NET 2021 [Question ID = 2349]

Given below are two statements :

Statement I: Cleanroom software process model incorporates the statistical quality certification of code increments as they accumulate into a system.

Statement II: Cleanroom software engineering follows the classic analysis, design, code, test, and debug cycle to software development and focussing on defect removal rather than defect prevention.

In light of the above statements, choose the correct answer from the options given below :

1. Both Statement I and Statement II are true.
2. Both Statement I and Statement II are false.
3. Statement I is true but Statement II is false.
4. Statement I is false but Statement II is true.

non-gatecse software-design software-reliability

Answer key

28.44.4 Software Reliability: UGC NET CSE | August 2016 | Part 2 | Question: 45

The extent to which a software performs its intended functions without failures, is termed as

A. Robustness
B. Correctness
C. Reliability
D. Accuracy

ugcnetcse-aug2016-paper2 is&software-engineering software-reliability

Answer key

28.44.5 Software Reliability: UGC NET CSE | December 2011 | Part 2 | Question: 16

Emergency fixes known as patches are result of

A. Adaptive maintenance
B. Perfective maintenance
C. Corrective maintenance
D. None of the above

ugcnetcse-dec2011-paper2 is&software-engineering software-reliability

Answer key

28.44.6 Software Reliability: UGC NET CSE | December 2014 | Part 3 | Question: 46

Maximum possible value of reliability is

A. 100
B. 10
C. 1
D. 0

ugcnetcse-dec2014-paper3 is&software-engineering software-reliability

Answer key

28.44.7 Software Reliability: UGC NET CSE | July 2016 | Part 3 | Question: 43

A server crashes on the average once in 30 days, that is, the Mean Time Between Failures (MTBF) is 30 days. When it happens it takes 12 hours to reboot it, that is, Mean Time To Repair (MTTR) is 12 hours. The availability of server with these reliability data values is approximately.

A. 96.3 %
B. 97.3 %
C. 98.3%
D. 99.3%

ugcnetcse-july2016-paper3 is&software-engineering software-reliability

Answer key

28.44.8 Software Reliability: UGC NET CSE | June 2013 | Part 3 | Question: 2

Match the following :

a.	Watson-Felix model	i.	Failure intensity
b.	Quick-Fix model	ii.	Cost estimation
c.	Putnam resouce allocation model	iii.	Project planning
d.	Logarithmetic Poission Model	iv.	Maintainace

Codes :

A. a-ii, b-i, c-iv, d-iii
B. a-i, b-ii, c-iv, d-iii
C. a-ii, b-i, c-iii, d-iv
D. a-ii, b-iv, c-iii, d-i

ugcnetcse-june2013-paper3 is&software-engineering software-reliability non-gatecse

Answer key

28.44.9 Software Reliability: UGC NET CSE | June 2013 | Part 3 | Question: 6

Software safety is a ______ activity that focuses on the identification and assessment of potential hazards that may affect software negatively and cause an entire system to fail.

A. Risk mitigation, monitoring and management
B. Software quality assurance
C. Software quality assurance
D. Defect removal efficiency

ugcnetcse-june2013-paper3 is&software-engineering software-reliability

Answer key

28.44.10 Software Reliability: UGC NET CSE | June 2014 | Part 3 | Question: 17

Assume that a program will experience 200 failures in infinite time. It has now experienced 100 failures. The initial failure intensity was 20 failures/CPU hr. Then the current failure intensity will be

A. 5 failured/CPU hr
B. 10 failured/CPU hr
C. 20 failured/CPU hr
D. 40 failured/CPU hr

ugcnetjune2014iii is&software-engineering software-reliability

Answer key

28.44.11 Software Reliability: UGC NET CSE | September 2013 | Part 3 | Question: 44

The failure intensity for a basic model as a function of failures experienced is given as $\lambda(\mu) - \lambda_0[1-(\mu)/(V_0)]$ where λ_0 is the initial failure intensity at the start of the execution, μ is the average or expected number of failures at a given point in time, the quantity V_0 is the total number of failures that would occur in infinite time.

Assume that a program will experience 100 failures in infinite time, the initial failure intensity was 10 failures/CPU hr. Then the decrement of failures intensity per failure will be

A. 10 per CPU hr
B. 0.1 per CPU hr
C. -0.1 per CPU hr
D. 90 per CPU hr

ugcnetcse-sep2013-paper3 is&software-engineering software-reliability

Answer key

28.45 Software Requirement Specification (1)

28.45.1 Software Requirement Specification: UGC NET CSE | September 2013 | Part 2 | Question: 31

Consider the following characteristics:

I. Correct and unambiguous
II. Complete and consistent
III. Ranked for importance and/or stability and verifiable
IV. Modifiable and Traceable

Which of the following is true for a good SRS?

A. I, II and III
B. I, III and IV
C. II, III and IV
D. I, II, III and IV

ugcnetsep2013ii is&software-engineering software-requirement-specification

Answer key

28.46 Software Requirement Specifications (1)

28.46.1 Software Requirement Specifications: UGC NET CSE | August 2016 | Part 2 | Question: 41

Which of the following is used to determine the specificity of requirements ?

A. $\frac{n_1}{n_2}$
B. $\frac{n_1}{n_2}$
C. $n_1 + n_2$
D. $n_1 - n_2$

Where n_1 is the number of requirements for which all reviewers have identical interpretations, n_2 is number of requirements in a specification.

ugcnetcse-aug2016-paper2 is&software-engineering software-requirement-specifications

Answer key

28.47 Software Reuse (1)

28.47.1 Software Reuse: UGC NET CSE | June 2019 | Part 2 | Question: 54

Software reuse is

A. the process of analysing software with the objective of recovering its design and specification
B. the process of using existing software artifacts and knowledge to build new software
C. concerned with reimplementing legacy system to make them more maintainable
D. the process of analysing software to create a representation of a higher level of abstraction and breaking software down into its parts to see how it works

ugcnetcse-june2019-paper2 software-reuse

Answer key

28.48 Software Testing (22)

28.48.1 Software Testing: Computer Science - UGC NET 2021 [Question ID = 2341]

In software testing, beta testing is the testing performed by _______________.

A. potential customers at the developer's location.
B. potential customers at their own locations.
C. product developers at the customer's location.
D. product developers at their own locations.

software-testing

Answer key

28.48.2 Software Testing: Computer Science - UGC NET 2021 [Question ID = 2343]

In software engineering, what kind of notation do formal methods predominantly use ?

1. Textual
2. Diagrammatic
3. Mathematical
4. Computer code

software-testing

Answer key

28.48.3 Software Testing: UGC NET CSE | August 2016 | Part 2 | Question: 44

For a program of k variables, boundary value analysis yields ______ test cases.

A. $4k-1$ B. $4k$ C. $4k+1$ D. 2^k-1

ugcnetcse-aug2016-paper2 is&software-engineering software-testing

Answer key

28.48.4 Software Testing: UGC NET CSE | December 2008 | Part 2 | Question: 43

Software Quality Assurance (SQA) encompasses :

A. verification
B. validation
C. both verification and validation
D. none of the above

ugcnetcse-dec2008-paper2 is&software-engineering software-testing

Answer key

28.48.5 Software Testing: UGC NET CSE | December 2009 | Part 2 | Question: 42

Any error whose cause cannot be identified anywhere within the software system is called ________

(A) Internal error

(B) External error

(C) Inherent error

(D) Logic error

ugcnetcse-dec2009-paper2 is&software-engineering software-testing

Answer key

28.48.6 Software Testing: UGC NET CSE | December 2009 | Part 2 | Question: 47

Black Box testing is done

(A) to show that s/w is operational at its interfaces i.e. input and output.
(B) to examine internal details of code.
(C) at client side.
(D) none of above.

ugcnetcse-dec2009-paper2 is&software-engineering software-testing

Answer key

28.48.7 Software Testing: UGC NET CSE | December 2010 | Part 2 | Question: 15

"Black" refers in the "Black-box" testing means

A. Characters of the movie "Black"
B. I – O is hidden
C. Design is hidden
D. Users are hidden

ugcnetcse-dec2010-paper2 is&software-engineering software-testing

Answer key

28.48.8 Software Testing: UGC NET CSE | December 2010 | Part 2 | Question: 41

Prototyping is used to

A. Test the software as an end product
B. Expand design details
C. Refine and establish requirements gathering
D. None of the above

ugcnetcse-dec2010-paper2 is&software-engineering software-testing

Answer key

28.48.9 Software Testing: UGC NET CSE | December 2011 | Part 2 | Question: 18

Following is used to demonstrate that the new release of software still performs the old one did by rerunning the old tests :

A. Functional testing
B. Path testing
C. Stress testing
D. Regression testing

ugcnetcse-dec2011-paper2 is&software-engineering software-testing

Answer key

28.48.10 Software Testing: UGC NET CSE | December 2012 | Part 2 | Question: 31

Basis path testing falls under

A. system testing
B. white box testing
C. black box testing
D. unit testing

ugcnetcse-dec2012-paper2 is&software-engineering software-testing

Answer key

28.48.11 Software Testing: UGC NET CSE | December 2013 | Part 3 | Question: 8

One weakness of boundary value analysis and equivalence partitioning is

A. they are not effective
B. they do not explore combinations of input circumstances
C. they explore combinations of input circumstances
D. none of the above

ugcnetcse-dec2013-paper3 is&software-engineering software-testing

Answer key

28.48.12 Software Testing: UGC NET CSE | December 2015 | Part 2 | Question: 46

In software testing, how the error, fault and failure are related to each other?

A. Error leads to failure but fault is not related to error and failure
B. Fault leads to failure but error is not related to fault and failure
C. Error leads to fault and fault leads to failure
D. Fault leads to error and error leads to failure

ugcnetcse-dec2015-paper2 is&software-engineering software-testing

Answer key

28.48.13 Software Testing: UGC NET CSE | January 2017 | Part 3 | Question: 43

Which of the following statement(s) is/are TRUE with regard to software testing?

I. Regression testing technique ensures that the software product runs correctly after the changes during maintenance.
II. Equivalence partitioning is a white-box testing technique that divides the input domain of a program into classes of data from which test cases can be derived.

A. only I
B. only II
C. both I and II
D. neither I nor II

ugcnetcse-jan2017-paper3 is&software-engineering software-testing

Answer key

28.48.14 Software Testing: UGC NET CSE | January 2017 | Part 3 | Question: 44

Which of the following are facts about a top-down software testing approach?

I. Top-down testing typically requires the tester to build method stubs
II. Top-down testing typically requires the tester to build test drivers

A. Only I
B. Only II
C. Both I and II
D. Neither I nor II

ugcnetcse-jan2017-paper3 is&software-engineering software-testing

Answer key

28.48.15 Software Testing: UGC NET CSE | June 2012 | Part 3 | Question: 30

While unit testing a module, it is found that for a set of test data, maximum 90% of the code alone were tested with a probability of success 0.9. The reliability of the module is

A. at least greater than 0.9
B. equal to 0.9
C. at most 0.81
D. at least 1/0.81

ugcnetcse-june2012-paper3 is&software-engineering software-testing

Answer key

28.48.16 Software Testing: UGC NET CSE | June 2013 | Part 3 | Question: 4

Equivalence partitioning is a ______ method that divides the input domain of a program into classes of data from which test cases can be derived

A. White-box testing
B. Black-box testing
C. Orthogonal array testing
D. Stress testing

ugcnetcse-june2013-paper3 is&software-engineering software-testing non-gatecse

Answer key

28.48.17 Software Testing: UGC NET CSE | June 2014 | Part 2 | Question: 15

Regression testing is primarily related to

A. Functional testing
B. Development testing
C. Data flow testing
D. Maintenance testing

ugcnetcse-june2014-paper2 is&software-engineering software-testing

Answer key

28.48.18 Software Testing: UGC NET CSE | June 2014 | Part 3 | Question: 16

Software testing is

A. the process of establishing that errors are not present
B. the process of establishing confidence that a program does what it is supposed to do
C. the process of executing a program to show that it is working as per specifications
D. the process of executing a program with the intent of finding errors

ugcnetjune2014iii is&software-engineering software-testing

Answer key

28.48.19 Software Testing: UGC NET CSE | Junet 2015 | Part 2 | Question: 42

Which of the testing strategy requirements established during requirements analysis are validated against developed software?

A. validation testing
B. integration testing
C. regression testing
D. system testing

ugcnetcse-june2015-paper2 software-testing is&software-engineering

Answer key

28.48.20 Software Testing: UGC NET CSE | September 2013 | Part 2 | Question: 30

Equivalence partitioning is a _____ testing method that divides the input domain of a program into classes of data from which test cases can be derived.

A. White box
B. Black box
C. Regression
D. Smoke

ugcnetsep2013ii is&software-engineering software-testing

Answer key

28.48.21 Software Testing: UGC NET CSE | September 2013 | Part 3 | Question: 29

____ refers to the discrepancy among a computed, observed or measured value and the true specified or theoretically correct values?

A. Fault
B. Failure
C. Defect
D. Error

ugcnetcse-sep2013-paper3 is&software-engineering software-testing

Answer key

28.48.22 Software Testing: UGC NET CSE | September 2013 | Part 3 | Question: 43

Equivalence class partitioning approach is used to divide the input domain into a set of equivalence classes, so that if a program works correctly for all the other values in that class. This is used _____

A. to partition the program in the form of classes
B. to reduce the number of test cases required
C. for designing the test cases in the white box testing
D. all of the above

ugcnetcse-sep2013-paper3 is&software-engineering software-testing

Answer key

28.49 Software Validation (1)

28.49.1 Software Validation: UGC NET CSE | December 2012 | Part 3 | Question: 65

_____ establishes information about when,why and by whom changes are made in a Software.

A. Software Configuration Management
B. Change Control
C. Version Control
D. An Audit Trail

ugcnetcse-dec2012-paper3 is&software-engineering software-validation

Answer key

28.50 Source Code Metric (1)

28.50.1 Source Code Metric: UGC NET CSE | December 2014 | Part 3 | Question: 43

Which one of the following is not a source code metric ?

A. Halstead metric
B. Function point metric
C. Complexity metric
D. Length metric

ugcnetcse-dec2014-paper3 is&software-engineering source-code-metric

Answer key

28.51 Uml (1)

28.51.1 Uml: UGC NET CSE | June 2013 | Part 3 | Question: 26

An actor in an animation is a small program invoked _____ per frame to determine the characteristics of some object in the animation.

A. once
B. twice
C. 30 times
D. 60 times

ugcnetcse-june2013-paper3 is&software-engineering uml

Answer key

28.52 Validation (1)

28.52.1 Validation: UGC NET CSE | June 2019 | Part 2 | Question: 60

Software validation mainly checks for inconsistencies between

A. use cases and user requirements
B. implementation and system design blueprints
C. detailed specifications and user requirements
D. functional specifications and use cases

ugcnetcse-june2019-paper2 validation

Answer key

28.53 Waterfall Model (2)

28.53.1 Waterfall Model: UGC NET CSE | August 2016 | Part 2 | Question: 42

The major shortcoming of waterfall model is

A. The difficulty in accommodating changes after requirement analysis.
B. The difficult in accommodating changes after feasibility analysis.
C. The system testing.
D. The maintenance of system.

ugcnetcse-aug2016-paper2 is&software-engineering waterfall-model

Answer key

28.53.2 Waterfall Model: UGC NET CSE | July 2016 | Part 2 | Question: 43

Which of the following is not included in the waterfall model?

A. Requirement analysis
B. Risk analysis
C. Design
D. Coding

ugcnetcse-july2016-paper2 waterfall-model

Answer key

28.54 White Box Testing (1)

28.54.1 White Box Testing: UGC NET CSE | June 2019 | Part 2 | Question: 53

In the context of software testing, which of the following statements is/are NOT correct?

P: A minimal test set that achieves 100% path coverage will also achieve 100% statement coverage.

Q: A minimal test set that achieves 100% path coverage will generally detect more faults than one that achieves 100% statement coverage.

R: A minimal test set that achieves 100% statement coverage will generally detect more faults than one that achieves 100% branch coverage.

A. R only
B. Q only
C. P and Q only
D. Q and R only

ugcnetcse-june2019-paper2 white-box-testing statement-coverage-branch-coverage-path-coverage

Answer key

Answer Keys

28.0.1	Q-Q	28.0.2	Q-Q	28.0.3	Q-Q	28.0.4	Q-Q	28.0.5	Q-Q
28.0.6	Q-Q	28.0.7	Q-Q	28.0.8	Q-Q	28.0.9	Q-Q	28.0.10	Q-Q
28.0.11	Q-Q	28.0.12	Q-Q	28.0.13	Q-Q	28.0.14	Q-Q	28.0.15	Q-Q
28.0.16	Q-Q	28.0.17	Q-Q	28.0.18	Q-Q	28.0.19	Q-Q	28.0.20	Q-Q
28.0.21	Q-Q	28.1.1	B	28.2.1	C	28.2.2	B	28.3.1	B
28.4.1	C	28.5.1	C	28.5.2	B	28.5.3	C	28.6.1	Q-Q
28.7.1	Q-Q	28.7.2	Q-Q	28.7.3	B	28.8.1	B	28.9.1	D
28.9.2	C	28.10.1	Q-Q	28.10.2	A	28.11.1	B	28.11.2	Q-Q
28.11.3	B	28.12.1	B	28.13.1	D	28.13.2	A	28.14.1	C
28.14.2	C	28.15.1	B	28.16.1	A	28.17.1	Q-Q	28.18.1	B
28.18.2	C	28.19.1	C	28.20.1	B	28.21.1	Q-Q	28.21.2	Q-Q
28.21.3	Q-Q	28.21.4	Q-Q	28.21.5	Q-Q	28.21.6	A	28.21.7	C
28.21.8	B	28.21.9	C	28.21.10	D	28.21.11	C	28.21.12	D
28.21.13	D	28.21.14	A	28.21.15	A	28.21.16	C	28.21.17	D
28.21.18	B	28.21.19	B	28.21.20	B	28.21.21	D	28.21.22	D
28.21.23	B	28.21.24	B	28.21.25	N/A	28.21.26	Q-Q	28.21.27	Q-Q
28.21.28	Q-Q	28.21.29	Q-Q	28.21.30	Q-Q	28.21.31	Q-Q	28.21.32	N/A
28.21.33	Q-Q	28.21.34	D	28.21.35	C	28.21.36	B	28.21.37	C
28.21.38	B	28.21.39	B	28.21.40	B	28.21.41	B	28.21.42	B

28.21.43	A	28.21.44	C	28.21.45	B	28.21.46	A	28.21.47	C
28.21.48	B	28.21.49	B	28.21.50	B	28.21.51	B	28.21.52	C
28.21.53	C	28.21.54	B	28.21.55	C	28.21.56	C	28.21.57	A
28.21.58	B	28.21.59	C	28.21.60	D	28.21.61	A	28.21.62	C
28.22.1	A	28.23.1	B	28.24.1	C	28.25.1	A	28.26.1	B
28.27.1	C	28.28.1	D	28.29.1	B	28.29.2	D	28.29.3	B
28.30.1	D	28.31.1	B	28.32.1	B	28.33.1	D	28.34.1	Q-Q
28.34.2	Q-Q	28.34.3	Q-Q	28.34.4	Q-Q	28.34.5	Q-Q	28.34.6	Q-Q
28.34.7	B	28.34.8	C	28.34.9	Q-Q	28.34.10	Q-Q	28.34.11	Q-Q
28.34.12	A	28.34.13	X	28.34.14	C	28.34.15	B	28.34.16	C
28.34.17	D	28.34.18	A	28.35.1	Q-Q	28.35.2	Q-Q	28.35.3	C
28.36.1	Q-Q	28.36.2	Q-Q	28.36.3	Q-Q	28.36.4	Q-Q	28.36.5	Q-Q
28.36.6	C	28.36.7	D	28.36.8	D	28.36.9	C	28.36.10	D
28.36.11	A	28.37.1	Q-Q	28.38.1	D	28.39.1	B	28.39.2	B
28.40.1	D	28.40.2	C	28.40.3	B	28.41.1	B	28.42.1	Q-Q
28.42.2	D	28.43.1	D	28.44.1	Q-Q	28.44.2	Q-Q	28.44.3	Q-Q
28.44.4	Q-Q	28.44.5	Q-Q	28.44.6	C	28.44.7	C	28.44.8	D
28.44.9	B	28.44.10	B	28.44.11	C	28.45.1	D	28.46.1	Q-Q
28.47.1	B	28.48.1	Q-Q	28.48.2	Q-Q	28.48.3	Q-Q	28.48.4	Q-Q
28.48.5	B	28.48.6	Q-Q	28.48.7	Q-Q	28.48.8	Q-Q	28.48.9	Q-Q
28.48.10	B	28.48.11	B	28.48.12	C	28.48.13	A	28.48.14	A
28.48.15	C	28.48.16	B	28.48.17	D	28.48.18	D	28.48.19	A
28.48.20	B	28.48.21	D	28.48.22	B	28.49.1	D	28.50.1	B
28.51.1	A	28.52.1	C	28.53.1	Q-Q	28.53.2	B	28.54.1	A